

Presentation

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Presentation Question

**Explain the difference between var,
let and const in JavaScript, When
Would you use each one?**

Introduction

JavaScript, var, let, and const are keywords used for declaring variables, but they differ in scope, mutability, and hoisting.

var is function-scoped and can be re-declared and re-assigned.

let is block-scoped and can be re-assigned but not re-declared.

const is also block-scoped and cannot be re-assigned or re-declared.

Introduction

var variables can be updated and re-declared within its scope;

let variables can be updated but not re-declared.

const variables can neither be updated nor re-declared.

They are all hoisted to the top of their scope.

But while **var** variables are initialized with undefined.

let and **const** variables are not initialized

Explanation

- **var is function-scoped and has been around since the early days of JavaScript.**
- **let is block-scoped and was introduced in ES6, offering a more predictable scoping mechanism.**
- **const is also block-scoped and is used to declare constants that cannot be reassigned after initialization.**



scope

- **VAR:**
Function-scoped (accessible within the entire function) or global-scoped (if declared outside any function).
- **LET:**
Block-scoped (accessible within the block where it's declared, including any nested blocks).
- **CONST:**
Block-scoped, same as let.

Hoisting:

var: Hoisted to the top of its scope (initialized with undefined), but not initialized.

let: Hoisted to the top of its scope, but not initialized.

Attempting to access it before declaration results in a "Temporal Dead Zone" error.

const: Hoisted to the top of its scope, but not initialized.

Attempting to access it before declaration results in a "Temporal Dead Zone" error.

When they can be use

- **var** primarily when working with legacy code or when you need function-level scope and the possibility of re-declaration and re-assignment. However, in modern JavaScript development, **let** and **const** are preferred due to their more predictable scoping.
- **let** for variables whose values might need to change during the program's execution, and when you need block-level scope.
- **const** for variables that represent constants or for values that should not be reassigned. It's a good practice to use **const** for variables that you don't intend to change, especially for objects and arrays where you want to ensure that the reference doesn't change, although you can still modify the contents of the object or array.

Summery

- **var, let, and const are three different ways to declare variables in JavaScript, each with its own scope and use cases**

Thank You

